2016 EXAMINATIONS

Lancaster University

Part II

COMPUTING AND COMMUNICATIONS [2.5 Hours]

SCC.311 Distributed Systems

Candidates are asked to answer THREE questions from FOUR; each question is worth a total of 25 marks.

Use a separate answer book for each question.

Question 1

1.a Explain carefully what the *client-server model* means and how the approach is used in the context of distributed systems.

Briefly discuss *four security vulnerabilities* in a client-server interaction.

[**4 marks**]

1.b Describe the main principles behind both *symmetric* and *asymmetric cryptography.* (4 marks) Compare and contrast the two approaches. (2 marks)
Note: you do not need to go into details of any particular cryptographic algorithms.

[**6 marks**]

1.c Assume two parties, Gordon and Zheng, want to communicate securely in a distributed system. More precisely, they want to be confident that they are both talking to each other (*mutual authentication*) and they want a session key to support future *secure communication* between them.

Needham and Schroeder proposed a 5-step protocol based on symmetric cryptography to achieve these aims:

i. Describe precisely what this 5-step protocol involves. (4 marks)

ii. Explain carefully how this protocol achieves mutual authentication, identifying the key steps involved in this process. (2 marks)

iii. This protocol relies on an authentication server to achieve the above properties. What important assumption is made about this service? (2 marks)

[**8 marks**]

1.d Suppose Gordon and Zheng decide they would rather use asymmetric cryptography:

i. Describe how asymmetric cryptography can be used to achieve a basic level of authentication, so that Gordon can be sure that Zheng is indeed Zheng (it is not necessary to show how mutual authentication is achieved). (1.5 marks)

ii. In addition, show how asymmetric approaches can achieve secure communication. (1.5 marks)

[**3 marks**]

Question 1 continues on next page...

Question 1 continued…

1.e Scalability is also a crucial concern in client-server computing:

i. What actions might you take at the server-end to enhance scalability? (2 marks)

ii. Similarly, are there any actions at the client-end that can add to the scalability of a design? (1 mark)

iii. To what extent can the security solution adopted influence the scalability of a design? (1 mark)

[**4 marks**]

Total 25 marks

Question 2

2.a Consider the following *definition* of a distributed system:

> "A distributed system is a system designed to *support* the development of applications and services which can *exploit* a physical architecture consisting of multiple, *autonomous* processing elements that *do not share primary memory* but *cooperate* by *sending asynchronous messages* over a communication network"

Discuss and dissect this quote, examining the significance of each individual phrase in turn.

[**4 marks**]

2.b The nature of the underlying communications channel, i.e. asynchronous and also potentially lossy, has significant repercussions for distributed systems in terms of the ability to achieve *consensus*. Discuss this problem in detail. You may wish to refer to the *Byzantine Generals Problem* to illustrate your answer.

[**6 marks**]

2.c Distributed systems must overcome a range of other problems including *partial failure* where different nodes may fail at different times, and also periods of *variable connectivity* or even *disconnection*. Describe what is meant by *indirect communication* and explain carefully how this paradigm might help with these problems.

[**4 marks**]

2.d *Publish-subscribe* is an example of an indirect communication service. Describe the main features of a publish-subscribe approach and distinguish between *content-based* and *topic-based* publish-subscribe. (4 marks)

Outline one approach that might enable you to implement a content-based solution efficiently in a *large-scale* distributed system. (2 marks)

Explain carefully how the strategy should scale. (1 mark)

[**7 marks**]

2.e *Message queues* provide an alternative indirect communication approach. Compare and contrast message queues with publish-subscribe.

[**4 marks**]

Total 25 marks

Question 3

3.a MapReduce stores the output produced by the Map function on the local disk of the Map worker. A Reduce worker that needs a Map worker's output then fetches the data over the network directly from the Map worker. In contrast, the Map inputs and Reduce outputs are stored in the GFS replicated distributed file system. Here we assume the GFS keeps at least two copies of the same file.

MapReduce could have been designed so that Map workers write Map output to GFS, and Reduce workers read the Map output from GFS. Briefly explain the **main performance disadvantages** and **fault-tolerance advantages** of switching from local disk to GFS for Map output.

[**4 marks**]

3.b Apache Zookeeper is an open source project, providing distributed configuration, synchronisation, and naming registry services for large-scale distributed systems. Passive replication is used by Zookeeper to coordinate replicated servers.

i. Define what passive replication is.

[**2 marks**]

ii. Briefly explain what are *the advantages and disadvantages* of passive replication compared to active replication in terms of *communication overhead, processing overhead, and recovery overhead*.

[**3 marks**]

iii. Ben has implemented an active replication scheme for Zookeeper. Assume Zookeeper runs in a *non-deterministic* environment and there are **two replicas** (so the replication overhead is low). Non-deterministic means replicas could give different answers for the same request, depending on how the operating system schedules them.

Will you use active replication in this environment? Briefly explain your decision.

[**3 marks**]

Question 3 continues on next page...

Question 3 continued…

3.c Assume you decide to implement a passive replication scheme and you come up with the following simplified pseudo code.

```
1  onReceive(request) {
2     if (I_am_the_primary) {
3          process request
4          send result to client
5          for each remaining backup replicas {
6               send update to the replica
7          }
8     } else {
9          Tell client who the primary is
10    }
11 }
```

In the following we assume duplicated requests are properly handled by the rest of your code, and that there is mechanism to safely switch to a new primary server (randomly chosen) if the primary fails, and have clients switch to this new primary. We also assume there is only one client connecting to your service.

i. Represent a failure free execution of the above code on a time-line diagram with four backup replicas.

[**3 marks**]

ii. After publishing your project on github (an open source repository), a developer suggested replacing the *for loop at line 5* with an atomic multicast operation.

Do you think this is necessary? (1 mark)
Justify your choice using an example, representing on a time-line diagram. (4 marks)

[**5 marks**]

iii. The developer decides to go ahead to create a branch of your project by replacing the *for loop at line 5* with atomic multicast. However, there is a problem if this change is made.

Describe a scenario in which the primary fails and the client receives an incorrect answer (1 mark). Illustrate your scenario on a time-line diagram (2 marks). Propose a modified version with pseudo-code to fix this problem while still using atomic multicast (2 marks). *Bear in mind that we assume there is only one client, and duplicated requests are properly handled.*

[**5 marks**]

Total 25 marks

Question 4

4.a Paxos is taught in the lecture on Google's infrastructure. In Paxos a consensus is reached when the majority of participants have agreed on a value (either accept or reject). Briefly explain why it would be a bad idea to use Paxos instead of two-phase commit to provide atomic distributed commit. **[5 marks]**

4.b What is a group communication service? (3 marks) What are the advantages of multicast over multiple unicasts? (2 marks).

[5 marks]

4.c Sarah has developed a voice-chat plugin for League of Legends -- a multiplayer online battle arena video game. Her plugin allows players of the same team to communicate with each other when they are playing the game.

Consider the following pseudo code of Sarah's implementation. Here ***send*** is a communication primitive which does not provide any delivery guarantee (i.e. using send, messages might be lost or reordered). ***multicast*** (line 7) sends a message to all players in the group. When a message arrives, the ***onReceive*** method (lines 6-9) *will be automatically invoked by the operating system*.

```
1  void voiceChat(voice) {
2          for each of player of the Group {
3                  send(voice);
4      }
5  }
6  void onRecive(voice, groupMembers) {
7          multicast(voice, groupMembers);
8          play the voice
9  }
```

i.

What type of reception mode does Sarah's plug-in implement? (1 mark)

What is the advantage of this reception mode? (2 marks)

What would be the alternative reception mode? (1 mark)

[4 marks]

Question 4 continues on next page...

Question 4 continued...

ii.

Sarah's implementation has a bug. What is the bug? (2 marks)

How will you fix the bug? Present your solution in pseudo code. (2 marks)

[**4 marks**]

iii.

Assume the bug got fixed. What level of reliability does Sarah's implementation provide? (1 mark)

What protection does this level of reliability provide? (1 mark)

[**2 marks**]

iv.

Consider the following voice messages received by Alice and Bob who happen to be in the same team.

Alice	Bob
"Hello, this is Bob"	"Let's play Summoner's Rift"
"Hello, this is Mike"	"Hello, this is Bob"
"Let's play Summoner's Rift"	"Hello, this is Josh"
"Hello, this is Josh"	"Hello, this is Alice"
"Hello, this is Alice"	

What kind of message ordering does the voice-chat plug-in provide here, if any? (1 mark)

What benefits and weaknesses do you see in this choice? (2 marks)

Give a possible reason why Bob did not receive "Hello, this is Mike" message? (1 mark)

Propose a potential countermeasure to ensure Bob receives this message. (1 mark)

[**5 marks**]

Total 25 marks

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